Biomimetic fish robots for underwater monitoring

"Review Scopus-indexed research (2020–2026) on biomimetic fish robots for low-disturbance underwater monitoring, integrating fish-inspired propulsion, ROS 2 control, PINN-based CFD, and soft robotics. Examine swimming efficiency, underwater noise, wake turbulence, and ecological impact. Identify existing integrated systems, experimental results, limitations, research gaps, and potential novelty. Include key papers, DOIs, and publication venues. - comprehensive analysis" *Generated by LeapSpace*

Biomimetic Fish Robots for Low-Disturbance Underwater Monitoring at a Glance

Key Findings Table

Theme	Key finding	Population / system type	Evidence
Integrated systems	No retrieved Scopus-indexed study evidences a single biomimetic fish robot that unifies fish-inspired propulsion, ROS 2 control, PINN-based CFD, and soft robotics for ecological monitoring; partial integration dominates instead [1] [2] [3] [4] [5] [6]	Biomimetic fish robots and adjacent underwater robots, 2020-2026	[1] [2] [3] [4] [5] [6]
Swimming efficiency	Efficiency gains recur when drive frequency, fin geometry, and stiffness are matched to structural resonance; intermediate or distributed stiffness usually outperforms rigid or overly soft designs [7] [8] [9] [10] [11] [12]	Caudal-fin, pectoral-fin, and soft/compliant fish robots	[7] [8] [9] [10] [11] [12]
Wake physics	Reverse Kármán vortices, vortex-ring timing, and schooling/wake matching are linked to thrust generation, higher efficiency, and lower transport cost [13] [14] [15] [16] [17]	Robotic fish propulsion and wake-interaction studies	[13] [14] [15] [16] [17]

Theme	Key finding	Population / system type	Evidence
ROS 2 adoption	ROS 2 appears only rarely in fish-inspired systems; advanced closed-loop control is much more common than middleware standardization [2] [18] [19] [20] [21]	Fish robots and underwater bioinspired robots	[2] [18] [19] [20] [21]
PINN-based CFD	Direct PINN application to fish-robot wake prediction or hydrodynamic optimization is absent in the retrieved set; the evidence points to an opportunity rather than an established practice [3] [4] [22] [23] [24]	Biomimetic fish robots and underwater CFD/ML studies	[3] [4] [22] [23] [24]
Soft robotics	Soft and compliant actuation improves adaptability and can support quiet undulation, but leakage, hysteresis, pressure sensitivity, fatigue, and sensor drift remain major field barriers [25] [26] [27] [28]	Soft robotic fish and underwater soft robots	[25] [26] [27] [28]
Disturbance and ecology	Low-disturbance claims are common, but direct acoustic, wake, and ecological-response measurements are sparse; most support is indirect or from narrow experiments [29] [30] [31] [32] [33] [34]	Monitoring-oriented fish robots, propeller comparisons, fish-response studies	[29] [30] [31] [32] [33] [34]

Direct Answer

The strongest evidence-supported conclusion is that biomimetic fish robots are technically promising for low-disturbance underwater monitoring, but the literature from 2020-2026 remains fragmented rather than fully integrated. Efficient swimming is best supported when compliant fins or bodies are tuned to resonance and distributed stiffness rather than simply driven at higher frequency [7] [8] [9] [10] [11]. Existing platforms show meaningful progress in three separate directions: fish-inspired propulsion with hydrodynamic validation [1] [35], modular/autonomous navigation including isolated ROS 2 use in an open-water batoid-inspired robot [2] [19], and soft/compliant robotic fish with measurable swimming gains and some endurance evidence [36] [37] [38]. However, no retrieved study demonstrates one Scopus-indexed system combining fish-inspired propulsion, ROS 2, PINN-based CFD, and soft robotics for ecological monitoring [1] [2] [3] [5] [6].

For low-disturbance monitoring specifically, the evidence is suggestive but not yet definitive. Biomimetic propulsion is repeatedly framed as quieter and less wake-disruptive than propellers [29] [32] [39], and there are some direct indicators: reduced TDS disturbance in a frog-like robot versus a propeller robot [40], lower-SPL propulsor selection in SABUVIS [32] [41], and fish-behavior studies showing robot design affects approach distance and tailbeat frequency in aquaculture settings [34]. But standardized hydroacoustic, PIV wake, and ecological-stress validation remains limited [31] [33] [42].

The clearest novelty opportunity is an end-to-end monitoring platform that combines a soft or variable-stiffness fish body, ROS 2-native autonomy, PINN- or surrogate-informed hydrodynamic/acoustic optimization, and ecological disturbance validation in real deployments [2] [6] [22] [23] [43].
Confidence level: High - this conclusion is explicitly supported by the meta-synthesis and repeated across multiple evidence streams [1] [2] [3] [5] [6].

Study Scope

- **Time window:** 2020-2026, as defined in the evidence base [3] [6] [19] [37] [44].
- **Indexing frame:** Scopus-indexed research, as specified in the query and evidence design [1] [45] [46] [47].
- **Disciplines covered:** biomimetic propulsion, marine robotics control/autonomy, CFD and ML surrogates, soft robotics, underwater monitoring, and ecological interaction [23] [47] [48] [49] [50].
- **Methods represented:** experiments, pool/open-water validation, CFD, FSI, PIV, dynamic modeling, closed-loop control, DRL/NMPC, and methodological PINN templates from adjacent fluid domains [1] [20] [23] [42] [51].

Assumptions & Limitations

- The supplied evidence does **not** provide a reliable, complete DOI-and-venue list for all key papers, so publication metadata can only be reported where titles/years were surfaced in the evidence text; identifiers are otherwise not supplied here to avoid fabrication [3] [47].
- Low-disturbance performance is often inferred from propulsion morphology or qualitative comparisons rather than standardized hydrophone, PIV, or animal-response protocols [29] [30] [32] [33].
- ROS 2 and PINN evidence comes partly from adjacent underwater robotics and fluid-modeling literature because direct fish-robot applications are sparse [3] [19] [23].

Suggested Further Research

- Build and field-test a **fully integrated** soft robotic fish with ROS 2 autonomy and surrogate-informed gait optimization [2] [23] [43].
- Establish a **disturbance benchmark suite** combining SPL, PIV wake intensity, suspended-solids change, and fish-behavior metrics [34] [40] [42].

- Apply **PINNs/HFM/XPINNs** to fish-robot wake reconstruction and low-noise gait optimization under sparse sensor data [22] [23] [24].
- Improve **durability and packaging** for long deployments, especially leakage resistance, pressure tolerance, and sensor-drift compensation [25] [26] [27].
- Standardize reporting of **cost of transport, Froude efficiency, acoustic signature, and wake metrics** for cross-paper comparison [32] [52] [53].

Full Report

1. Foundational Context

Biomimetic fish robots aim to reproduce body/caudal-fin (BCF) or median/paired-fin (MPF) propulsion to achieve efficient, agile, and potentially low-disturbance underwater locomotion [48]. In this literature, performance is usually judged with metrics such as swimming speed, thrust, cost of transport, Froude efficiency, turning radius, and sometimes Strouhal-number-dependent wake behavior [53] [54] [55]. The central hydrodynamic idea is that efficient propulsion emerges from favorable vortex formation rather than continuous screw-driven momentum injection, which is why reverse Kármán wakes and resonance-matched fin motion recur as design targets [10] [13] [14].

For monitoring applications, “low disturbance” can refer to at least four different things: reduced wake turbulence, reduced sound pressure or propulsor noise, reduced sediment or water-quality perturbation, and reduced behavioral disturbance to aquatic organisms [30] [33] [34] [40]. The literature often conflates these, so a careful review must separate hydrodynamic, acoustic, and ecological disturbance claims [32] [47].

Confidence level: High - the conceptual framework is consistently supported across propulsion, monitoring, and review papers [47] [48] [55].

2. Scope, Search Logic, and Evidence Base

The evidence base covers Scopus-indexed work from 2020-2026 structured around four technical pillars: propulsion efficiency and wake physics, autonomy/control with emphasis on ROS 2, PINN-related CFD surrogates, and soft robotics, with environmental monitoring as the application lens [1] [45] [46] [47]. The retrieved studies include experiments, reviews, simulation-validation papers, and adjacent underwater-robot systems where they directly inform integration or deployment readiness [1] [6] [19] [50] [56].

A major synthesis result is that the field is technically rich but systemically fragmented: propulsion-softness integration is common, control-sensing integration is increasingly sophisticated, and CFD/ML methods are growing nearby, yet full convergence is not evidenced [1] [2] [3] [5] [6].

Confidence level: High - this is directly supported by the meta-analysis and repeated topic findings [1] [2] [3] [5] [6].

3. Fish-Inspired Propulsion, Hydrodynamic Efficiency, and Wake Physics

3.1 Efficiency design rules

Across the evidence, the most reproducible design rule is not “higher tail-beat frequency equals better performance,” but “match actuation to stiffness and natural frequency” [7] [8] [10]. Experimental and FSI studies show that medium or intermediate flexibility commonly improves thrust and/or efficiency over rigid fins, while very soft fins can lose thrust even if efficiency rises in some conditions [7] [57] [58]. A 2024 trailing-edge-shape study found distinct optimal bands relative to natural frequency, with square and deeply forked fins each favored in different reduced-frequency ranges [7]. A 2024 stiffness-optimization study similarly identified best performance at intermediate stiffness and frequency combinations [8]. Decreasing stiffness from head to tail also improved performance in optimization studies [9] [59].

These findings align with broader soft-robotics evidence showing stiffness adaptation can roughly double efficiency and produce energy savings relative to fixed configurations [11]. Taken together, these findings suggest that variable-stiffness tails are one of the most mature routes to higher-performance low-disturbance swimming [8] [11].
Confidence level: High - multiple independent studies point to the same rule [7] [8] [9] [11] [12].

3.2 Experimental performance

Measured performance spans several actuation classes and body plans. Reported examples include a highest Froude efficiency of 0.84 at a particular amplitude/frequency condition in one biomimetic fish [54], a multi-flexible robotic fish reaching 1.63 BL/s at 3 Hz with low energy cost [60], a piezoelectric robotic trout reaching 0.84 BL/s in still water and 0.71 BL/s under imposed flow [36], and a DEA-based soft fish reaching 0.25 BL/s near its modeled natural frequency [61]. Decoupling propulsion and steering yielded up to 50% higher cruising speed and 30-40% lower cost of transport than a coupled baseline in one study [62]. A hydraulic variable-stiffness fish raised average speed by 118% and reduced turning radius to roughly one quarter in another [37].

Cost-of-transport reporting remains inconsistent; one review explicitly notes COT is seldom reported in fish robotics, limiting rigorous cross-platform comparison [52].
Confidence level: Medium-High - the performance evidence is direct, but metrics are heterogeneous [36] [37] [52] [60].

3.3 Wake turbulence and vortices

Wake structure is not merely descriptive; it is increasingly treated as a design variable. Reverse Kármán vortices were linked to jet generation and positive force, yielding high Froude efficiency and outperforming traditional propeller propulsion in one analysis [13]. For a pectoral-fin robot, thrust peaks tracked vortex-ring formation and detachment, implying transient thrust enhancement depends on unsteady shedding timing [14]. Schooling studies indicate leader-follower spacing and phase can preserve useful vortices, reduce cost of transport, and increase thrust and efficiency [15]. An artificial lateral-line study targeted turbulence-noise suppression and reduced error by 36% after adding flow-control components [17].

This pattern implies future monitoring robots could use wake-aware control both to minimize disturbance and to exploit ambient vortices for efficiency [15] [17].
Confidence level: Medium - strong mechanistic support exists, but monitoring-specific transfer is still mostly inferred [13] [14] [15] [17].

4. ROS 2, Onboard Autonomy, and Underwater Navigation

Explicit ROS 2 adoption in fish-inspired systems is rare. One retrieval pass found no papers explicitly mentioning ROS 2 across the fish-robot autonomy set [18]. More targeted retrieval identified a ray-inspired underwater robot with ROS 2-based modular software for closed-loop attitude, depth, and heading control, validated in open water with waypoint navigation and intermittent GPS surfacing fixes [19]. A 2026 batoid-inspired platform is the clearest fish-like example explicitly using ROS 2 for onboard closed-loop navigation [2].

By contrast, sophisticated closed-loop control without ROS 2 is common. Demonstrated methods include CPG with onboard infrared and IMU sensing for obstacle avoidance [63], semi-autonomous navigation with 0.12 m/s average speed, 0.8 s response time, and about 5% navigation error [64], vision-based CPG/PID/fuzzy control with 0.65 m minimum turning radius [65], and NMPC with disturbance observation reducing RMSE to 1.73 m in multi-obstacle simulations while keeping outdoor position errors within 0.2 body lengths [20]. DRL, DDPG, and Q-learning are also used for obstacle avoidance and path planning [66] [67] [68].

The overall pattern is clear: control sophistication is rising faster than software standardization [2] [20] [21].
Confidence level: High - ROS 2 scarcity and control-method richness are both directly evidenced [2] [18] [19] [20].

5. PINN-Based CFD, Surrogate Modeling, and Design Optimization

The retrieved evidence does not show direct application of PINNs or PINN variants to fish-robot CFD, wake prediction, or hydrodynamic design optimization [3] [4]. Fish-robot hydrodynamics remain dominated by classical CFD, immersed-boundary simulation, FSI, and data-assisted dynamic models [69] [70].

The closest transferable work lies in adjacent domains. Underwater-vehicle studies report CFD+ML surrogates such as a physics-informed GAN for AUV hydrodynamic performance [56], an adapted ResNet-50 flow-field predictor [71], and a DNN hull-drag surrogate with 1.85% MAPE and about 100x speedup [72]. Fish-related control studies also use surrogate environments or physics-guided models, but not PINNs specifically for fish wake optimization [73] [74].

Methodologically, the strongest templates for future fish-robot work are HFM for noisy sparse wake visualization [22], NSFnets for incompressible Navier-Stokes recovery including hidden pressure [23], and XPINNs for complex irregular geometries and space-time decomposition [24]. These are well aligned with robotic-fish problems involving deforming bodies, partial flow sensing, and many-query optimization [75] [76].

So PINN-based fish hydrodynamics is best described as an open novelty space rather than an established line of work [3] [23].
Confidence level: High - the absence of direct evidence is explicit, and adjacent templates are well documented [3] [4] [22] [23] [24].

6. Soft Robotics, Compliant Actuation, and Durability Limits

Soft robotic fish span pneumatic/fluidic, hydraulic, SMA, DEA, HASEL, MFC, IPMC, and hybrid architectures [38] [77] [78] [79]. Their main advantages are lifelike undulation, safer interaction, and sometimes silent or low-noise motion [80] [81]. Performance benchmarks include the piezoelectric trout at 0.84 BL/s [36], DEA fish at 0.25 BL/s [61], and hydraulic systems capable of extended operation up to 36 h in one architecture [82].

Hybrid or variable-stiffness designs are especially promising because they broaden operating range and improve the thrust-efficiency Pareto frontier [5] [43] [83]. Reconfigurable tails with lockable joints and layer jamming extend gait diversity across anguilliform to thunniform modes [84]. Yet many stiffness-changing mechanisms remain bulky, low-bandwidth, or anatomically unrealistic [85] [86] [87].

Durability remains a core bottleneck. Reported issues include fatigue microdamage leading to leaks [26], water absorption and hysteresis [25], hydrostatic-pressure degradation [27], sensor drift and nonlinear calibration problems [25], and internal micro-bubbles that weaken deep-sea elastomer reliability [28]. Proposed mitigations include strain-distributing geometries, sealing/encapsulation, reinforcement layers, defect screening, self-healing or fatigue-tolerant materials, and continuous calibration [25] [28] [88] [89].
Confidence level: High - the trade-off between adaptability and fieldability is consistently documented [25] [26] [27] [28].

7. Low-Disturbance Monitoring, Noise, Wake Disturbance, and Ecological Effects

7.1 Deployment evidence

Actual monitoring-oriented deployment is limited but real. RoboSea-2.0 was used in fish ponds for multipoint, multidepth water-quality monitoring with stable data transmission and real-time underwater movement [90]. UJIFISH-I was validated for aquaculture inspection and monitoring, reaching 20 m depth with long-range remote operation and reliable target detection while using undulatory propulsion intended to reduce turbulence and disturbance [6].

7.2 Noise and disturbance

The strongest direct disturbance experiment in the provided set is from a bionic frog robot: TDS changed only 0.17% from still water versus 1.48% for a propeller robot during water-quality testing [40]. SABUVIS compared propulsor-drive types using generated-noise measurements to choose the lower-SPL option, though the numeric difference is not surfaced in the evidence [32] [41]. A 2025 biomimetic propeller study reported up to 6.67 dB noise reduction versus the original propeller through serrated/microstructured blade designs [31]. A 2025 fin-propulsion study used water-tunnel tests and PIV with low-hydroacoustic-noise intent [42].

Still, most “quiet” claims remain comparative or theoretical, rooted in known propeller noise sources such as blade rotation, vortex shedding, and stern-wake pressure fluctuations [30] [91] [92].
Confidence level: Medium - there is direct evidence, but it is sparse and not standardized [31] [32] [40] [41].

7.3 Ecological interaction

Behavioral-response evidence is narrow but important. In a large aquaculture sea cage, salmon were more likely to swim close to a flipper-propelled robotic fish and showed lower tailbeat frequency; robot type affected behavior [34]. In another study, repeated conspecific-like robotic-fish approaches increased avoidance, with freezing and evasive maneuvers dependent on distance and relative speed [33]. These results show that “biomimetic” does not automatically mean ecologically neutral [33] [34].

For monitoring design, this implies disturbance must be validated behaviorally, not assumed from morphology alone [93] [94].
Confidence level: Medium-High - the studies are direct, but the evidence base is small [33] [34].

8. Existing Integrated Systems, Key Papers, and Metadata Status

Most relevant integrated or near-integrated papers surfaced in the evidence

Paper / title as surfaced	Year	What it contributes	Venue / DOI status in supplied evidence
A continuum soft robotic trout with embedded HASEL actuators: design, fabrication, and swimming kinematics	2024	Soft continuum body and embedded actuation; useful for soft integration [95]	Venue/DOI not surfaced [95]
Trout-like multifunctional piezoelectric robotic fish and energy harvester	2021	Measured swimming speed benchmark for compliant actuation [36]	Venue/DOI not surfaced [36]
Undulatory Swimming Performance Explored With a Biorobotic Fish and Measured by Soft Sensors and Particle Image Velocimetry	2021	PIV-based hydrodynamic validation [51]	Venue/DOI not surfaced [51]
Ray-inspired underwater robot with ROS2 closed-loop control	not fully titled in evidence	Explicit ROS 2 onboard control and open-water waypoint tracking [19]	Venue/DOI not surfaced [19]
2026 batoid-inspired underwater locomotion paper	2026	Modular ROS2 software for closed-loop navigation [2]	Venue/DOI not surfaced [2]
UJIFISH-I	not dated in excerpt	Aquaculture monitoring deployment and modular sensing [6] [94]	Venue/DOI not surfaced [6] [94]
RoboSea-2.0	not dated in excerpt	Fish-pond water-quality monitoring deployment [90]	Venue/DOI not surfaced [90]
Fish-like aquatic propulsion studied using a pneumatically-actuated soft-robotic model	2020	Soft propulsion thrust characterization [96]	Venue/DOI not surfaced [96]

The evidence is adequate for identifying key papers by title or platform name, but not for a verified DOI-and-venue catalogue.
Confidence level: High - this limitation is explicit in the supplied material [3] [47].

9. Main Limitations, Gaps, and Potential Novelty

The dominant gaps are consistent across topics:

1. **No fully integrated platform** combining fish-inspired propulsion, ROS 2, PINN-based CFD, and soft robotics for monitoring [1] [2] [3] [5] [6].
2. **ROS 2 scarcity** despite advanced control methods [2] [18] [19].
3. **No direct PINN fish-hydrodynamics practice** in the retrieved set [3] [4].
4. **Sparse standardized low-disturbance validation** across acoustics, wake, and ecology [31] [34] [40].
5. **Soft-robot durability bottlenecks** for long deployments [25] [26] [27].

A strong novelty claim, grounded in these gaps, would be a **ROS 2-native soft robotic fish with online variable stiffness and a PINN/surrogate wake-acoustics model**, validated in real monitoring tasks against a propeller baseline using SPL, PIV wake intensity, suspended-solids change, and fish-behavior assays [2] [22] [23] [34] [40] [43].
Confidence level: High - the novelty follows directly from explicit absences and adjacent enabling evidence [2] [3] [6] [23] [43].

10. Conclusion

The 2020-2026 Scopus-indexed literature shows real momentum in biomimetic fish robots for low-disturbance monitoring, especially in resonance-aware propulsion design, compliant actuation, and increasingly advanced closed-loop control [7] [8] [11] [20]. But the field is not yet integrated: ROS 2 remains exceptional, PINN-based fish CFD is largely absent, and ecological low-disturbance claims are much stronger than the direct measurement base [2] [3] [32]. The most defensible research opportunity is therefore not another isolated propulsion paper, but a unified field platform that couples soft fish-like mechanics, modular autonomy, physics-informed flow optimization, and direct disturbance validation [2] [6] [23] [43].

Overall confidence level: High - the central conclusion is strongly supported by convergent evidence across all four technical pillars and the monitoring literature [1] [2] [3] [5] [6].

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